Math Inquirers

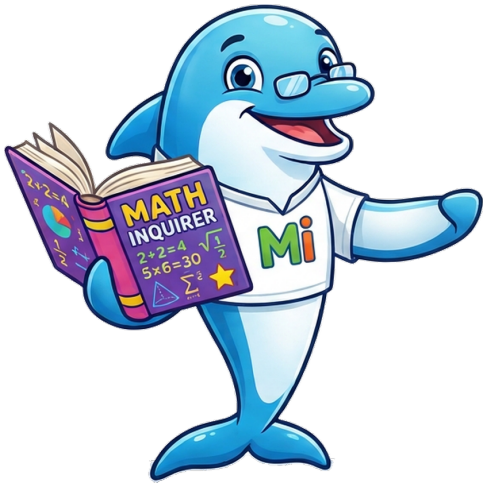
Think. Question. Prove.

Level 1 (ages 6-8) - Term 1

Tutor Answer Key

Book 2

Lessons 6 to 10



FOR THE TUTOR AND GROWN-UPS ONLY - keep away from the child

Contents

Math Inquirers Level 1, Term 1 - Tutor Answer Key, Book 2

Lesson 6 **Tens Jump, Ones Hop** page 3
Hop

Lesson 7 **Take Away, Count Up** page 7
Addition And Subtraction

Lesson 8 **Balance and Swap** page 11
Addition And Subtraction: Balance

Lesson 9 **Draw and Measure** page 15
Length: Draw And Measure

Lesson 10 **Story Steps: Read It Twice** page 19
Story Steps

Answer key: Lesson 6, Tens Jump, Ones Hop

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.

Answer: 3 bundles, 7 singles, 37 sticks

The picture shows 2 bundles and 7 singles, which is 27. One more bundle makes 3 bundles, and the 7 singles stay. 30 and 7 is 37.

Watch for: Adding the bundle to the singles (28).; Forgetting the old bundles (17).

S2. In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?

Answer: 54 (5 bars drawn and 4 singles)

3 bars and 4 singles is 34. Two more bars add 20, so there are 5 bars and 4 singles: 54.

Watch for: Adding the 2 to the singles (36).; Drawing 2 sticks instead of 2 bars of ten.

S3. Jump forward. Fill in the blanks.

Answer: 33, 63, 81, 67

(a) 2 tens and 1 ten make 3 tens: 33. (b) 2 tens and 4 tens make 6 tens: 63. (c) 5 tens and 3 tens make 8 tens: 81. (d) 7 is 7 ones; add 6 tens to get 67.

Watch for: Changing the ones digit (23 + 40 = 27).; (d) writing 13 (7 + 6).

S4. Jump or hop. Fill in the blanks.

Answer: 24, 38, 55, 69

(a) 3 tens take away 1 ten is 2 tens: 24. (b) 5 tens take away 2 tens is 3 tens: 38. (c) 3 is a hop: 8 ones take away 3 is 5 ones: 55. (d) 7 is a hop forward: 2 and 7 make 9 ones: 69.

Watch for: Taking the 3 from the tens digit (58 - 3 = 28).; Hopping the wrong way in (d).

S5. Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?

Answer: 38 pencils

3 boxes of ten is 30 pencils. 68 - 30 = 38: jump back 3 tens (58, 48, 38). The 8 ones stay.

Watch for: Taking away 3 (65).; Adding instead of taking away (98).

S6. Ben says 45 + 20 = 47. Fill in a to c.

Answer: No. The right answer is 65. He added 2 ones instead of 2 tens.

20 is 2 tens, so Ben should jump two tens: 45, 55, 65. He added 2 ones and got 47. A jump changes the tens digit.

Watch for: Saying he is correct because 20 has a 2 in it.; Not saying what he mixed up (any clear sentence about tens and ones is fine).

S7. Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?

Answer: 56 (two jumps of ten from 36: 46, then 56)

36 + 10 = 46, then 46 + 10 = 56. Two jumps of ten land on 56. The 6 ones never change.

Watch for: Drawing 2 small hops (38).; Changing the ones digit.

Think harder (page 3)

T1. Add 23 + 45. Fill in the blanks.

Answer: 68 (jump 40 to 63, then hop 5 to 68)

45 is 4 tens and 5 ones. 23 + 40 = 63. Then 63 + 5 = 68. Check: 2 tens and 4 tens are 6 tens; 3 ones and 5 ones are 8 ones.

Watch for: Adding only the tens and stopping at 63.; Adding 4 + 5 as 9 tens (93).

T2. Subtract 79 - 35. Fill in the blanks.

Answer: 44 (jump back 30 to 49, then hop back 5 to 44)

35 is 3 tens and 5 ones. 79 - 30 = 49. Then 49 - 5 = 44.

Watch for: Taking 35 away from the ones only.; Hopping forward instead of back.

T3. Both frogs start on 12. Frog A jumps 3 tens, then hops 4. Frog B hops 4, then jumps 3 tens. Fill in a to c.

Answer: Yes, both land on 46 (30 and 4 are added in either order)

Frog A: 12 + 30 = 42, then 42 + 4 = 46. Frog B: 12 + 4 = 16, then 16 + 30 = 46. They land on the same number: adding the same jumps and hops in a different order gives the same total.

Watch for: Saying Frog B lands on 16 because the hops came first.; Adding 3 + 4 and landing on 19.

T4. Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?

Answer: 48 (jump back 3 tens from 78)

Undo the 3 jumps of ten: 78, 68, 58, 48. Check: 48 + 30 = 78.

Watch for: Adding 30 to 78 (108).; Taking away only 3 (75).

Go further (page 4)

G1. Mei is on 7 and must get to 97. Each time she can jump 10 or jump 20. Many answers are right. Fill in a and b.

Answer: Any two different routes of 10s and 20s from 7 to 97, e.g. 7, 17, 27, 37, 47, 57, 67, 77, 87, 97 (nine jumps of 10) and 7, 27, 47, 67, 87, 97 (four of 20 and one of 10).

Every jump adds 10 or 20, and all the jumps together add 90 (97 - 7). Examples: nine jumps of 10; or 20, 20, 20, 20, 10 landing on 27, 47, 67, 87, 97. The ones digit stays 7 on every landing.

Watch for: Landing on a number that is not 10 or 20 more than the one before.; Two routes that are the same jumps in the same order.; Stopping at 87 or going past 97.

G2. Now Mei wants the fewest jumps from 7 to 97. She can jump 10 or jump 20. Fill in a and b.

Answer: 5 jumps: four jumps of 20 and one jump of 10 (any order). Four jumps travel at most 80, so she reaches only 87.

She must travel 9 tens. Four jumps of 20 make 8 tens. One jump of 10 makes the last ten. That is 5 jumps. Four jumps can travel at most 80, which only reaches 87, so 5 is the fewest.

Watch for: Saying 4 (80 only reaches 87).; Saying 9 (only using jumps of 10).

G3. Start at 25. Jump forward 3 tens, jump back 1 ten, then jump forward 2 tens. Then hop and land on 69. Fill in a to c.

Answer: 55, 45, 65, then a hop of 4 to 69. The three jumps are the same as 4 tens forward.

25 + 30 = 55. 55 - 10 = 45. 45 + 20 = 65. To land on 69 the hop is 4. Short way: 3 - 1 + 2 = 4 tens forward, so 25 + 40 = 65.

Watch for: Jumping forward on the jump back.; Saying the hop is 44 (69 - 25).; Saying 6 tens for (c) (adding 3 + 1 + 2).

Investigation (page 5): Jump road

From 6 you land on 16, 26, 36, 46, 56 and so on: every number ends in 6, so you never land on 50. Jumps of ten never change the ones digit. Only a start number that ends in 0 (0, 10, 20, 30, 40 or 50 itself) can land on 50.

Look for: A child who tests more than one start number, says the ones digit stays the same, and gives a reason using tens and ones (a jump is whole bars). A child who says start 20 can land on 50 because 20 + 30 = 50.

Exit ticket (page 5)

E1. Add.

Answer: 76

4 tens and 3 tens make 7 tens. The 6 ones stay: 76.

Watch for: Writing 49 (adding 3 to the ones).; Writing 16.

E2. Subtract.

Answer: 44 and 71

(a) 30 is 3 tens: jump back to 44. (b) 3 is a hop: hop back to 71.

Watch for: Doing the same for both parts.; (b) writing 44.

E3. Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.

Answer: 72 and 75

2 tens forward from 52 is 72. Then a hop of 3 forward is 75.

Watch for: Writing 54 (adding 2 to the ones).; Stopping at 72.

Sneak-peek puzzles (not in the workbook)

1. 36
2. No. 43 + 20 = 63. 20 is 2 tens, so the tens digit changes, not the ones.

Tens Jump, Ones Hop: home practice answers

Name _____ Date _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 75
4 tens and 3 tens make 7 tens. The 5 loose pencils stay: 75.

2. 46 and 82
40 is 4 tens, a jump back: 46. 4 is a hop back: 82.

3. Ravi 50, Mei 23
Ravi hops back 3: 50. Mei jumps back 3 tens: 23.

4. 59 both ways. Yes, the same.
Jump first: 36 + 20 = 56, then + 3 = 59. Hop first: 36 + 3 = 39, then + 20 = 59.

5. 7, 17, 27 … 97. The ones digit stays the same.
A grown-up checks that each number is 10 more than the one before. The ones digit never changes when you jump in tens.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 7, Take Away, Count Up

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Here is 68 as bars and squares. Cross out 2 bars. Then fill in the blanks.

Answer: 4 tens and 8 ones are left; 68 - 20 = 48

68 is 6 tens and 8 ones. Take away 2 tens: 4 tens and 8 ones are left. That is 48. The ones did not change.

Watch for: Crossing out 2 squares instead of 2 bars (66).; Writing 84 or 4 + 8 = 12.

S2. Subtract. Use your kit if you like.

Answer: 37, 53, 56, 81

(a) 5 tens take away 2 tens is 3 tens: 37. (b) 7 ones take away 4 ones is 3 ones: 53. (c) 8 tens take away 3 tens is 5 tens: 56. (d) 6 ones take away 5 is 1: 81.

Watch for: Changing the wrong digit (57 - 20 = 55).; Taking away from both digits.

S3. Find 68 - 25. Fill in the blanks.

Answer: 48, then 43

68 take away 20 is 48. Then 48 take away 5 is 43. So 68 - 25 = 43.

Watch for: Taking away 25 as 2 + 5 = 7 (61).; Forgetting to start part (b) from 48.

S4. Subtract. Then add to check.

Answer: 34, then 46 (the same as the start, so it checks)

46 take away 10 is 36, and 36 take away 2 is 34. Check: 34 + 12 = 46, the number we started with. So 46 - 12 = 34 is right.

Watch for: Not adding back the number taken away (adding 46 + 12).; Writing 58 for the check.

S5. Mei has 45 stickers. She gives away 13. How many stickers are left? Draw a bar for each ten and a square for each one.

Answer: 32 (4 bars and 5 squares drawn; 1 bar and 3 squares crossed out)

45 is 4 tens and 5 ones. 13 is 1 ten and 3 ones. 45 take away 10 is 35. 35 take away 3 is 32. So Mei has 32 left.

Watch for: Crossing out 13 squares.; Writing 58 (adding instead of taking away).

S6. Kumar says 59 - 20 = 57.

Answer: No. 39. He took away 2 ones instead of 2 tens (57 is 59 - 2).

59 is 5 tens and 9 ones. Take away 2 tens: 3 tens and 9 ones, which is 39. Kumar took away 2 ones, so he got 57.

Watch for: Saying he is right.; Writing 79 (adding 20).

S7. A bus has 64 seats. 21 children are sitting. How many seats are empty?

Answer: 43 seats

64 take away 20 is 44. Then 44 take away 1 is 43. So 43 seats are empty.

Watch for: Adding 64 + 21 = 85.; Taking away only 2 or only 1.

Think harder (page 3)

T1. Mei has 62 beads. Ravi has 38 beads. Find how many more beads Mei has than Ravi.

Answer: 24 more beads (38 to 40 is 2, 40 to 60 is 20, 60 to 62 is 2)

38 to 40 is 2. 40 to 60 is 20. 60 to 62 is 2. The whole gap is 2 + 20 + 2 = 24. Check: 38 + 24 = 62.

Watch for: Forgetting that two tens is 20 (writing 2).; Writing 62 - 38 as 36 or 34.; Subtracting the smaller digit from the larger in each column (62 - 38 = 36).

T2. Find 83 - 77. Show how you found it.

Answer: 6 (77 to 80 is 3, 80 to 83 is 3)

77 to 80 is 3. 80 to 83 is 3. The gap is 3 + 3 = 6. Check: 77 + 6 = 83.

Watch for: Writing only one hop (3).; Taking away 83 - 7 = 76.; Subtracting the smaller digit from the larger in each column (83 - 77 = 14).

T3. Ali has 24 stickers and Ben has 31 stickers. Then each child gets 10 more stickers. Find how many more stickers Ben has than Ali now.

Answer: 7 more (the gap stays the same: 34 and 41, and 41 - 34 = 7)

Ali now has 34 and Ben has 41. The gap from 34 to 41 is 7. It was 7 before too (24 to 31), because both children got the same 10 more.

Watch for: Writing 17 (7 + 10).; Writing 41 (Ben's number).

T4. A book has 71 pages. Ravi has read 46 pages. How many pages are left?

Answer: 25 pages (46 to 50 is 4, 50 to 70 is 20, 70 to 71 is 1)

46 to 50 is 4. 50 to 70 is 20. 70 to 71 is 1. The gap is 4 + 20 + 1 = 25. Check: 46 + 25 = 71.

Watch for: Adding 71 + 46.; Forgetting the last hop of 1 (24).; Subtracting the smaller digit from the larger in each column (71 - 46 = 35).

Go further (page 4)

G1. Fill in the blanks. Circle the quicker way for each.

Answer: 71 - 68 = 3 (count up: the numbers are close). 71 - 20 = 51 (take away: 2 tens).

71 - 68: count up. 68 to 70 is 2 and 70 to 71 is 1, so 3. The numbers are close, so the gap is short. 71 - 20: take away 2 tens to get 5 tens and 1 one, which is 51. Counting up 20 would be a long way.

Watch for: Taking away 68 from 71 one at a time.; Counting up from 20 to 71 (too long).

G2. A crate has 100 eggs. Ali sells 64 of them. How many eggs are left? Find the answer two ways.

Answer: 36 eggs. Take away: 100 - 60 = 40, then 40 - 4 = 36. Count up: 64 to 70 is 6, then 70 to 100 is 30, so 36. They match.

Take away: 100 - 60 = 40, and 40 - 4 = 36. Count up: 64 to 70 is 6, and 70 to 100 is 30, so 6 + 30 = 36. Both ways give 36. Check: 64 + 36 = 100.

Watch for: Writing 164 (adding).; Writing 40 (forgetting to take away the 4 ones).

G3. Mei says 83 - 47 and 86 - 50 have the same answer.

Answer: Yes. Both are 36. Both numbers go up by 3, so the gap does not change.

83 - 47: 47 to 50 is 3, 50 to 80 is 30, 80 to 83 is 3, so 36. 86 - 50 = 36. 83 became 86 and 47 became 50: both went up by 3, so the gap between them stayed 36.

Watch for: Finding only one answer.; Saying the answers differ because the numbers differ.

Investigation (page 5): When is counting up quicker?

Counting up is quicker when the two numbers are close together, so the gap is short (52 - 49 is 3). Taking away is quicker when the number taken away is small or a whole number of tens (52 - 10 is 42). If the numbers are far apart and the small number is awkward, both ways take longer.

Look for: A child who tries several pairs, says what they notice about how far apart the numbers are, and gives a reason. Accept any clear rule that matches examples. A child who also checks by adding is doing well.

Exit ticket (page 5)

E1. Subtract.

Answer: 34

58 take away 20 is 38, and 38 take away 4 is 34.

Watch for: Writing 82 (adding).; Writing 36 (took away 2 and 2).

E2. Fill in the blanks.

Answer: 5, then 5

57 to 60 is 3, and 60 to 62 is 2. That is 5. So 62 - 57 = 5.

Watch for: Writing only the first hop (3).; Writing 15 (62 - 47).

E3. Find 90 - 87. Circle the way you used: take away or count up.

Answer: 3. Either way is fine; counting up from 87 to 90 is short because the numbers are close together.

87 to 90 is 3, so 90 - 87 = 3. Counting up is quick because the numbers are close.

Watch for: Taking away 87 from 90 one at a time.; Writing 13.

Sneak-peek puzzles (not in the workbook)

1. 11
2. 4 more. From 58 to 60 is 2, and from 60 to 62 is 2 more. 2 and 2 make 4.

Take Away, Count Up: home practice answers

Name _____ Date _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 58
78 is 7 tens and 8 ones. Take away 2 tens: 5 tens and 8 ones, which is 58.

2. 33; 33 + 43 = 76
76 take away 40 is 36, and 36 take away 3 is 33. Check: 33 + 43 = 76.

3. 53
89 take away 30 is 59. 59 take away 6 is 53.

4. Ali has 24 more (47 to 50 is 3, 50 to 70 is 20, 70 to 71 is 1)
47 to 50 is 3. 50 to 70 is 20. 70 to 71 is 1. 3 + 20 + 1 = 24.

5. For 38 and 71: 71 - 38 = 33, and counting up gives 2 + 30 + 1 = 33. Both ways match.
A grown-up checks that the child adds the hops and jumps, then adds the answer back to the smaller page number to get the greater one.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 8, Balance and Swap

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Fill in the missing numbers.

Answer: 4, 5, 3, 2

(a) 5 + 3 is 8 and 4 + 4 is 8. (b) 6 + 4 is 10 and 5 + 5 is 10. (c) 7 + 3 is 10 and 3 + 7 is 10. (d) 2 + 8 is 10 and 8 + 2 is 10.

Watch for: Writing the total of the left side in the blank (8 in part a).; Thinking the equals sign means the answer comes next.

S2. Mei puts 6 buttons and 2 buttons on the left plate. On the right plate she puts 7 buttons and some more. Draw the right plate in the box so both plates are the same.

Answer: 1 more button (6 + 2 = 8 and 7 + 1 = 8)

The left plate has 6 + 2 = 8 buttons. The right plate has 7 buttons, so it needs 1 more to make 8.

Watch for: Drawing 8 more buttons on the right plate.; Forgetting the 7 buttons already there.

S3. Siti writes 6 + 3 = 5 + 9.

Answer: No. The blank should be 4 (6 + 3 = 9 and 5 + 4 = 9). Siti wrote the total of the left side, as if = meant the answer comes next.

6 + 3 is 9. The right side must also be 9. 5 and 4 more make 9, so the blank is 4. Siti wrote 9, the total of the left side, but the blank is only one of the two numbers on the right.

Watch for: Saying yes because 9 is the answer to 6 + 3.; Writing 9 in the box again (copying Siti).

S4. 7 + 5 = 12. Fill in the missing numbers.

Answer: 6, 7, 8

(a) 6 + 6 = 12. (b) 5 + 7 = 12. (c) 4 + 8 = 12. The first number goes down by 1 and the second goes up by 1, so the total stays 12.

Watch for: Leaving the second number as 5.; Making the second number go down as well.

S5. Fill in the missing numbers.

Answer: 10, 10, 30, 40

(a) 20 went up by 20 to 40, so 30 goes down by 20 to 10. (b) 50 went up by 10 to 60, so 20 goes down by 10 to 10. (c) 40 went down by 10 to 30, so 20 goes up by 10 to 30. (d) 30 went down by 10 to 20, so 30 goes up by 10 to 40.

Watch for: Writing the total of the left side (50, 70, 60 or 60).; Moving both numbers the same way (part c: 20 - 10 = 10).

S6. Mei has 9 red pens and 6 blue pens. Ben has 10 red pens and some blue pens. They have the same number of pens. How many blue pens does Ben have?

Answer: 5 blue pens (9 + 6 = 10 + 5)

Mei has 9 + 6 = 15 pens. Ben has 10 red pens, so he needs 5 more to make 15. Another way: his red pens are 1 more than Mei's, so his blue pens are 1 fewer: 5.

Watch for: Answering 6 (copying Mei's blue pens).; Answering 15 (the total).

S7. Fill in the missing number.

Answer: 45 (35 + 12 = 47 and 45 + 2 = 47)

The 12 went down by 10 to become 2, so the 35 goes up by 10 to become 45. Check: 35 + 12 = 47 and 45 + 2 = 47.

Watch for: Writing 47 (the total).; Writing 25 (taking a ten from the 35).

Think harder (page 3)

T1. Mei says 56 + 17 and 54 + 19 have the same total. Ali says 54 + 19 has the greater total. Do not add.

Answer: Mei. 56 went down by 2 to 54 and 17 went up by 2 to 19. The 2 moved across, so both totals are 73.

56 to 54 is 2 less. 17 to 19 is 2 more. The 2 moved from one number to the other, so the total stays the same. Both sums make 73.

Watch for: Saying 54 + 19 is greater because 19 is a greater number than 17.; Adding both sums and not saying how the numbers moved (correct, but check the reason).

T2. Do not add. Circle LEFT or RIGHT for the greater total, or circle SAME.

Answer: SAME, SAME, RIGHT

(a) 29 up 1 to 30, 14 down 1 to 13: SAME (both 43). (b) 53 down 3 to 50, 20 up 3 to 23: SAME (both 73). (c) 26 up 2 to 28 and 15 up 1 to 16: both went up, so RIGHT is bigger (41 and 44).

Watch for: Saying SAME for part c because both numbers changed.; Looking only at the biggest numbers.

T3. Fill in the missing number. Then find the total.

Answer: 25, then the total is 75

49 went up by 1 to 50, so 26 goes down by 1 to 25. 50 + 25 = 75, so 49 + 26 = 75.

Watch for: Writing 27 (making 26 go up as well).; Writing 75 in the blank.

T4. Ben writes 24 + 17 = 26 + 19.

Answer: No. 24 + 17 = 41 but 26 + 19 = 45. Both numbers went up by 2. Change 19 to 15.

24 went up by 2 to 26, so 17 should go down by 2 to 15. Ben made it go up by 2 to 19, so the right side is 4 too big. Check: 24 + 17 = 41 and 26 + 15 = 41.

Watch for: Saying it is balanced because both numbers changed by 2.; Changing the 26 and not saying which number you change.

Go further (page 4)

G1. Fill in the missing numbers so that all three sums have the same total.

Answer: 27 and 17 (every sum is 47)

19 up 1 to 20, so 28 down 1 to 27. The total is 20 + 27 = 47 (check: 19 + 28 = 47). Then [] + 30 = 47, so the second blank is 47 - 30 = 17.

Watch for: Writing 27 in both blanks.; Writing 30 or 28 as the second blank.

G2. Fill in the blanks so both sides are the same. Write 3 different pairs. Many answers are right.

Answer: Any pair where the first blank is 4 more than the second, for example 5 and 1, 6 and 2, 7 and 3 (also 4 and 0).

If the second blank is 1, the right side is 10, so the first blank is 5. If it is 2, the right side is 11, so the first blank is 6. The first blank is always 4 more than the second, because 9 is 4 more than 5.

Watch for: Writing the same number in both blanks.; Giving one pair only.

G3. Mei has 28 stickers on a page and 17 in her pocket. Ali, Ben and Kumar have the same total as Mei. Fill in the blanks.

Answer: 15, 20 and 33 (Mei has 45 stickers; 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45)

Mei has 28 + 17 = 45. Ali: 30 is 2 more than 28, so his pocket has 2 fewer than 17: 15. Ben: 25 is 3 fewer than 28, so his pocket has 3 more than 17: 20. Kumar: his pocket 12 is 5 fewer than 17, so his page has 5 more than 28: 33. Check: 30 + 15 = 45, 25 + 20 = 45 and 33 + 12 = 45.

Watch for: Writing 17 for Ali's pocket.; Making Ben's pocket 14 (moving the wrong way).; Writing 17 or 45 for Kumar's page.

Investigation (page 5): Keep it the same

First step: 7 + 5 is not 10 (the totals are 10, 10 and 12). From 6 + 4 to 7 + 3 the 4 goes down 1. Next: add some to one number and take the same from the other and the total stays the same (give and take). Add 3 to both: the total goes up by 6. Take 2 from both: the total goes down by 4. When both numbers change the same way, the total changes by double.

Look for: A child who tries more than one pair of numbers, says it stays the same when one goes up and the other goes down by the same amount, and notices that changing both numbers the same way changes the total by double. Any clear reason is fine.

Exit ticket (page 5)

E1. Fill in the missing number.

Answer: 4

9 went up by 1 to 10, so 5 goes down by 1 to 4. Both sides make 14.

Watch for: Writing 14 (the total).; Writing 6 (making the 5 go up).

E2. Is 28 + 6 = 30 + 4 balanced?

Answer: Yes

28 went up by 2 to 30 and 6 went down by 2 to 4. Both sides make 34.

Watch for: Writing no because the numbers look different.

E3. Fill in the missing number.

Answer: 2. The 30 went up by 10 to 40, so the 12 goes down by 10 to 2.

30 went up by 10, so 12 goes down by 10 to 2. Both sides make 42.

Watch for: Writing 42 (the total).; Writing 22 (making the 12 go up).

Sneak-peek puzzles (not in the workbook)

1. Yes, the same: 10 beads each.
2. 4. 9 is 1 more than 8, so the blank is 1 less than 5. (Both sides make 13.)

Balance and Swap: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 4
6 + 3 is 9, and 5 + 4 is 9. The 6 became 5 (1 less), so the 3 becomes 4 (1 more).

2. 5
10 went down by 1 to 9, so 4 goes up by 1 to 5. Both sides make 14.

3. 10
20 went up by 20 to 40, so 30 goes down by 20 to 10. Both sides make 50.

4. Yes. 36 went down by 2 to 34 and 14 went up by 2 to 16. Both make 50.
One number goes down by 2 and the other goes up by 2, so the total stays the same.

5. Any list that shows the first number going down by 1 and the second going up by 1, for example 7 + 5, 6 + 6, 5 + 7, 4 + 8, 3 + 9, 2 + 10, 1 + 11, 0 + 12.
Each time one thing moves across, the total stays 12. A grown-up can count the things on both piles to check.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 9, Draw and Measure

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. In the box, draw a line 7 cm long. Start at the 0 mark.

Answer: The line ends at the 7 mark (it starts at the 0 mark and is 7 cm long)

Put a dot for the start. Line up the 0 mark with the dot. Make a small mark at the 7. Join the dot and the mark with a straight line. The line ends at the 7 mark, so it is 7 cm long.

Watch for: Starting at the edge of the ruler instead of the 0 mark (the line comes out a little too long).; Drawing to the 8 mark because the 0 was counted as one.

S2. In the box, draw a line 9 cm long. Ali's line is 4 cm long.

Answer: 5 cm longer (9 take away 4)

Your line is 9 cm and Ali's line is 4 cm. Count on from 4 to 9: that is 5 jumps. So your line is 5 cm longer (9 take away 4 is 5).

Watch for: Writing 13 by adding the two lengths.; Starting the 9 cm line at a mark other than 0.; Writing 9 (the length of the line you drew).

S3. Measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.

Answer: Any answer that matches the thing you measured and is close to your guess. A grown-up checks.

A grown-up checks: put one end on the 0 mark and read the mark at the other end. A crayon is about 8 to 9 cm and a rubber about 5 to 6 cm. Accept any whole number within 1 cm of the tip.

Watch for: Starting at the edge of the ruler instead of the 0 mark.; Writing the guess in the measure space, or leaving off the unit.

S4. Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.

Answer: 13 cm; 9 cm; Ali's is 4 cm longer

Both start at the 0 mark, so each tip mark is the length: Ali 13 cm, Mei 9 cm. From 9 to 13 is 4 jumps (13 take away 9 is 4), so Ali's pencil is 4 cm longer.

Watch for: Adding 13 + 9 = 22.; Writing 9 as the difference (that is the length of Mei's pencil).

S5. Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.

Answer: The second line is 8 cm long (5 + 3)

Mei's line is 5 cm. 3 cm longer is 5 + 3 = 8 cm. Draw it from the 0 mark to the 8 mark.

Watch for: Drawing a line 3 cm long.; Starting the line at the edge of the ruler, so it is a little too long.

S6. Look at Mei's line on the ruler. Mei says it is 8 cm long.

Answer: No. Her line is 7 cm. She should draw on to the 8 mark

Mei started at 0 and stopped at the 7 mark, so her line is 7 cm long. A line 8 cm long must end at the 8 mark. She should draw on to the 8 mark.

Watch for: Saying yes because Mei said 8.; Counting the 0 as one, so the 7 mark looked like 8.

Think harder (page 3)

T1. Use your ruler to draw a path with 3 straight pieces: 5 cm, 3 cm and 4 cm.

Answer: 12 cm (5 + 3 + 4)

Draw 5 cm, turn, draw 3 cm, turn, draw 4 cm. The path is as long as all three pieces together: 5 + 3 = 8, and 8 + 4 = 12 cm.

Watch for: Measuring in a straight line from the start to the end of the path.; Drawing one long line instead of turning at the corners.

T2. Look at line A on the ruler. Line B starts at the 3 mark and is 6 cm long.

Answer: 7 cm; the 9 mark

(a) From the 2 mark to the 9 mark is 7 jumps: 9 take away 2 is 7 cm, not 9 cm. (b) Start at the 3 mark and jump on 6: 3 + 6 = 9. The line ends at the 9 mark.

Watch for: Writing 9 for line A because the end is at the 9 mark.; Taking 6 away from 3, or writing 6 for line B.

T3. Look at the ruler. Both pencils start at the 0 mark. Each tip is between two marks.

Answer: 9 cm; 13 cm; 4 cm longer

Ali's tip is just past the 9 mark, so it is closer to 9: 9 cm. Mei's tip is just before the 13 mark, so it is closer to 13: 13 cm. From 9 to 13 is 4 jumps, so Mei's pencil is 4 cm longer.

Watch for: Choosing the mark the tip has not reached (12 for Mei).; Writing 10 for Ali because the tip looks nearly there.; Writing 3 by using 12 and 9.

Go further (page 4)

G1. A ribbon is 30 cm long. Mei cuts all of it into 3 pieces: A, B and C. Piece A is 8 cm long. Piece B is 3 cm longer than piece A. How long is piece C?

Answer: Piece C is 11 cm (B is 11 cm, A and B together are 19 cm, and 30 take away 19 is 11)

B is 8 + 3 = 11 cm. A and B together are 8 + 11 = 19 cm. The ribbon is 30 cm, so C is 30 take away 19, which is 11 cm. Check: 8 + 11 + 11 = 30.

Watch for: Taking 30 - 8 = 22 and stopping.; Writing 3 for piece B instead of 3 more than A.

G2. A snail crawls along a long ruler from the 4 mark to the 13 mark. Then it crawls back 5 cm.

Answer: At the 8 mark; it crawled 14 cm in all (9 cm forward and 5 cm back)

Forward from the 4 mark to the 13 mark is 9 cm. Back 5 cm: 13 take away 5 puts the snail at the 8 mark. In all it crawled 9 + 5 = 14 cm, although it is only 4 cm from where it began.

Watch for: Writing 9 for the total and forgetting the way back.; Writing 4 as the mark (the distance from the start, not the mark).

G3. Draw a path with 3 straight pieces. The pieces are all different lengths, in whole cm. The whole path is exactly 20 cm long. Many answers are right.

Answer: Many answers are right: any 3 different whole numbers that add up to 20, for example 5, 7 and 8 (5 + 7 + 8 = 20).

One answer is 5, 7 and 8: 5 + 7 is 12 and 12 + 8 is 20. Other answers are 3, 8 and 9, or 4, 7 and 9. If the total is not 20, change one piece until it is.

Watch for: Two pieces the same length (for example 5, 5 and 10).; A total of 19 or 21 because a piece was drawn a little too long or too short.

Investigation (page 5): Does the start mark matter?

The length stays 6 cm every time. From the 2 mark the line ends at 8; from 3 it ends at 9; from 5 it ends at 11. The end mark is always the start mark plus 6, so the end mark take away the start mark is always 6. Reading only the end mark gives the length only when the line starts at 0.

Look for: A child who tries more than one start mark, writes both marks, and says that the end mark take away the start mark (or the number of jumps) is the length. A 15 cm ruler allows start marks up to 9.

Exit ticket (page 5)

E1. In the box, draw a line 6 cm long. Start at the 0 mark.

Answer: A line from the 0 mark to the 6 mark

Put the 0 mark of the ruler on the dot and draw to the 6 mark. The line ends at the 6 mark.

Watch for: Starting at the edge of the ruler.; Ending at the 7 mark after counting the 0 as one.

E2. Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.

Answer: 9 cm longer

The rubber is 6 cm and the pencil is 15 cm. 15 take away 6 is 9, so the pencil is 9 cm longer.

Watch for: Adding: 21.; Writing 15 (the pencil, not the difference).

E3. A line starts at the 3 mark and ends at the 10 mark. How long is the line?

Answer: 7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)

Count the jumps from 3 to 10: 7 jumps. Or take 10 take away 3, which is 7. The line is 7 cm long, not 10 cm.

Watch for: Writing 10 (the end mark).; Writing 13 by adding.

Sneak-peek puzzles (not in the workbook)

1. A line from the dot that is 3 cm long: the 0 mark on the dot and the 3 mark at the other end.
2. No. It is 8 cm long. The pencil starts at the 1 mark, not the 0 mark, so the first centimetre of the ruler is not part of the pencil (9 take away 1 is 8).

Draw and Measure: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. It starts at 0 and ends at 8
A line 8 cm long, drawn from the 0 mark, ends at the 8 mark.

2. 6 cm
The pencil is 17 cm. 17 take away 11 is 6 (count on from 11 to 17: 6 jumps).

3. Many answers are right. Check one by measuring again.
Put one end at the 0 mark and read the mark at the other end. If the end is between two marks, either mark next to it is fine. Then put the numbers in order, smallest first.

4. 11 cm (15 take away 4)
Count the jumps from 4 to 15: 11 jumps. Or take 4 from 15, which is 11. It is not 15 cm, because the line starts at the 4 mark.

5. Many answers are right: any 3 different whole numbers that add up to 25, for example 6 + 8 + 11 = 25.
Choose three different numbers that add to 25, draw each piece with the 0 mark at the start, then measure and add to check.

More practice online (no timers, no scores): ask your tutor for the practice link.

Answer key: Lesson 10, Story Steps: Read It Twice

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Ben has 30 cards. He gives 20 cards away. After that, he gets 6 more cards.

Answer: Step 1: 30 - 20 = 10. Step 2: 10 + 6 = 16. Ben has 16 cards.

First, Ben gives away 20, so we take away: 30 - 20 = 10. Then he gets 6 more, so we add: 10 + 6 = 16. Ben has 16 cards now.

Watch for: Adding the 20 (30 + 20) because 20 is in the story.; Using 30 again in Step 2 instead of the 10 left after Step 1.

S2. Mei has 42 stickers. She gets 7 more stickers. Then she gives 10 stickers to Ali. Write one number sentence for each step.

Answer: Step 1: 42 + 7 = 49. Step 2: 49 - 10 = 39. Mei has 39 stickers.

Step 1: she gets 7 more, so add: 42 + 7 = 49. Step 2: she gives 10 away, so take away: 49 - 10 = 39. Mei has 39 stickers now.

Watch for: Using 42 again in Step 2 (42 - 10 = 32).; Adding both numbers (42 + 7 + 10).

S3. Ravi has 25 pencils. He buys 10 more pencils. Then he loses 3 pencils. Circle the 2 number sentences that fit the story.

Answer: Circle 25 + 10 = 35 and 35 - 3 = 32. Ravi has 32 pencils.

Step 1: buys 10 more, so 25 + 10 = 35. Step 2: loses 3, so 35 - 3 = 32. The sentences 25 - 10 = 15 and 35 + 3 = 38 use the wrong sign.

Watch for: Circling 25 - 3 = 22 because both numbers are in the story.; Circling a sentence with the wrong sign.

S4. Write + or - in each box.

Answer: +, -, -, +

Finds more means add (+). Eats means take away (-). Gives away means take away (-). Join means more, so add (+).

Watch for: Choosing + for gives away because there is a number in the sentence.; Choosing - for join.

S5. Ali has 23 marbles. Ben has 10 more than Ali. How many marbles do they have altogether? Use counters if you like.

Answer: Step 1: Ben has 23 + 10 = 33. Step 2: 23 + 33 = 56. They have 56 marbles.

Step 1: Ben has 10 more than Ali, so 23 + 10 = 33. Step 2: altogether means add Ali and Ben: 23 + 33 = 56 marbles.

Watch for: Answering 33 (stopping after Step 1).; Adding 23 + 10 and calling that altogether.

S6. A bus has 30 children. 10 children get off. Then 4 children get on. Siti writes 30 + 10 = 40 for Step 1.

Answer: No. Get off means take away: 30 - 10 = 20. Then 20 + 4 = 24. There are 24 children.

Children who get off leave the bus, so Step 1 is a take away: 30 - 10 = 20. Siti added the 10. Then 4 get on, so add: 20 + 4 = 24.

Watch for: Circling Yes.; Adding both numbers to 30 (30 + 10 + 4 = 44).

Think harder (page 3)

T1. Ravi has 15 marbles. He has 3 more marbles than Mei.

Answer: Mei has 12 marbles. Together they have 27.

Ravi has 3 more than Mei, so Mei has 3 fewer than Ravi: 15 - 3 = 12. Together: 15 + 12 = 27 marbles.

Watch for: Adding 15 + 3 = 18 because the word more is there.; Answering 12 for the second part.

T2. Mei is in a line to buy lunch. There are 6 children in front of her and 4 children behind her. How many children are in the line?

Answer: 11 children (6 + 4 + 1, counting Mei).

The children in front are 6 and behind are 4: 6 + 4 = 10. Mei is in the line as well, so 10 + 1 = 11 children.

Watch for: Answering 10 and leaving Mei out.

T3. Ben has 12 marbles. Mei has 5 more marbles than Ben. Siti has 3 fewer marbles than Mei. How many marbles does Siti have?

Answer: Mei has 17. Siti has 14 marbles.

Step 1: Mei has 5 more than Ben: 12 + 5 = 17. Step 2: Siti has 3 fewer than Mei: 17 - 3 = 14 marbles.

Watch for: Taking 3 from 12 (Siti is compared with Mei, not Ben).; Answering 17 (stopping after Step 1).

T4. Ali has 12 stickers. He gives 7 stickers to Mei. Then he gives 8 stickers to Ben.

Answer: 5 stickers left; No, the story cannot happen.

Step 1: 12 - 7 = 5, so Ali has 5 stickers left. Step 2 needs 8 stickers, but 5 is less than 8. Ali cannot give Ben 8, so this story can not be true. A story step can be impossible, so check each step.

Watch for: Writing 12 - 7 - 8 and getting an answer without noticing 5 is less than 8.; Circling Yes without checking step 2.

Go further (page 4)

G1. A shop has 60 pencils. It sells 24 pencils on Monday and 15 pencils on Tuesday. On Wednesday, a box of 30 pencils comes in. Write one number sentence for each step.

Answer: Step 1: 60 - 24 = 36. Step 2: 36 - 15 = 21. Step 3: 21 + 30 = 51. The shop has 51 pencils.

Monday: 60 - 24 = 36. Tuesday: 36 - 15 = 21. Wednesday a box comes in, so add: 21 + 30 = 51 pencils.

Watch for: Adding 24 and 15 and taking that from 60 but forgetting the box.; Starting each step from 60 again.

G2. Step 1: 25 + 10 = 35. Step 2: 35 - 8 = 27. Write a story that fits both steps. End your story with a question. Many stories are right.

Answer: Any story where 25 grows by 10 and then 8 are taken away, and the question asks how many now. The answer is 27.

For example: Ali has 25 stickers. He gets 10 more. Then he gives 8 to his sister. How many stickers does Ali have now? 25 + 10 = 35, 35 - 8 = 27.

Watch for: A story where the second step starts from 25 instead of 35.; No question at the end.

G3. Mei, Ali and Ben have 70 shells altogether. Mei has 25 shells. Ali has 10 more shells than Mei. Write one number sentence for each step. How many shells does Ben have?

Answer: Ali has 35. Mei and Ali have 60. Ben has 10 shells.

Step 1: Ali has 25 + 10 = 35. Step 2: Mei and Ali have 25 + 35 = 60. Step 3: Ben has the rest: 70 - 60 = 10 shells.

Watch for: Answering 35 or 60 (stopping early).; Taking only Mei's shells from 70.

Investigation (page 5): Does the order matter?

Both orders give the same answer (20 + 10 − 5 = 25 and 20 − 5 + 10 = 25) whenever both orders can be done. But the order can matter for what you are allowed to do: from 4 you cannot take away 6 first, though 4 + 10 − 6 = 8 works. So: the answer is the same, but you cannot take away more than you have.

Look for: A child who tests several numbers, says what they notice, and finds a case where taking away first is not possible (start 4, take away 6).

Exit ticket (page 5)

E1. A bus has 25 children. 10 children get off. Then 3 children get on. How many children are on the bus now?

Answer: 18 (25 − 10 = 15, then 15 + 3 = 18)

Get off means take away: 25 − 10 = 15. Get on means add: 15 + 3 = 18.

Watch for: Writing 38 (adding the 10).; Writing 15 (stopping after Step 1).

E2. Ben has 6 stickers. Mei has 5 more stickers than Ben. How many stickers does Mei have?

Answer: 11 (6 + 5)

Mei has more than Ben, so add: 6 + 5 = 11.

Watch for: Writing 1 (taking away).; Writing 5.

E3. Siti has 20 beads. Ali has 7 fewer beads than Siti. How many beads does Ali have?

Answer: 13 (20 − 7)

Ali has fewer, so take away: 20 − 7 = 13.

Watch for: Writing 27 (adding because the number 7 is there).; Writing 7.

Sneak-peek puzzles (not in the workbook)

1. 7
2. 11. First find Ben's marbles: 4 + 3 = 7. Then add Ali's 4: 4 + 7 = 11.

Story Steps: Read It Twice: home practice answers

Name _____ **Date** _____

For grown-ups. Please let your child try first, then check together. Ask *how do you know?* If your child is stuck, wait 10 seconds, then suggest drawing it.

1. 33
Step 1: 28 + 10 = 38. Step 2: 38 - 5 = 33.

2. 20
Siti has more, so add: 14 + 6 = 20.

3. 10
Take away the red and the blue: 40 - 10 = 30, then 30 - 20 = 10.

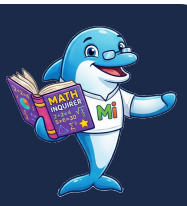
4. 22
Ben has fewer, so Kumar has more: 9 + 4 = 13. Together: 9 + 13 = 22.

5. Any two-step story with the right sentences. Check each step: the second sentence starts from the answer to the first.
A grown-up checks that each step has one add or one take away, and that the answer sentence matches the question.

More practice online (no timers, no scores): ask your tutor for the practice link.

TUTOR NOTES • LEVEL 1 • TERM 1 • MI'S CHALLENGE QUEST

Mi's Challenge Quest



8 gates, one from each lesson of the term, all at Start here level, plus one optional secret cave item. 20 minutes after the break, played in pairs or as a group quest with Mi, working on private paper. It is a game: no score or mark is shown to the child, no ranking, no visible timer. You keep private notes only. For the tutor: do not hand these notes to families.

Where it sits. Lesson 10 (the last lesson of the term), minutes 63 to 83, after the break and the workbook; or at the make-up slot. The last slot of the term is a game with Mi, not a test. The group plays it in pairs (a child and a quest partner) or as a group quest with Mi, on private paper. No score or mark is shown to the child, there is no ranking and no visible timer. The tutor keeps private notes only, for one line to each grown-up and for advice for next term's group. It keeps the 20-minute slot after the break and the question quality: 8 short gates and one optional secret cave item, all at Start here level. Lessons 2 and 4 share gate 2 and Lessons 6 and 7 share gate 5, so every lesson of the term is still visited. A middle child needs about 16 minutes of reading aloud and quiet work; a slower child may use the 5 flex minutes at the end.

1. The gates at a glance

Gate	From	Area (tutor label)	What it uses (Start here level)	Answer
1	L1	Whole Numbers	Tens and ones: read a picture of sticks in tens and ones	**3 tens and 9 ones; 39 sticks**
2	L2	Whole Numbers	Number line (the number the arrow shows) and counting on in tens (Lessons 2 and 4)	**67; 44**
3	L3	Whole Numbers	Make the greatest number from two cards	**83**
4	L5	Addition and Subtraction	Add by making a ten, then with a tens number	**12; 42**
5	L6	Addition and Subtraction	Jump in tens (52 + 30) and take away tens and ones (74 - 31) (Lessons 6 and 7)	**82; 43**
6	L8	Addition and Subtraction	Equals as a balance: a box number	**6; 20**
7	L9	Length; Addition and Subtraction in context	Draw a line of a given length, then find a length 4 cm longer	**a line 6 cm long; 10 cm**
8	L10	Length; Addition and Subtraction in context	A two-step story, one operation per step	**35 + 10 = 45; 45 - 3 = 42 (Mei has 42 stickers)**
Secret cave	L10	Length; Addition and Subtraction in context	Reading trap: who is in the group (optional)	**9 children (5 + 3 + 1, counting Ali)**

The area labels are for the tutor only and never appear on a child or parent page. Each gate is a fresh example of the Start here idea of its lesson (numbers chosen so that they do not repeat a lesson page), so a child who followed the lesson can open it. Before first use, read each gate against what the lesson actually taught.

2. Playing it: before and during

1. A week before: send the grown-up note (L10_Quest_Parent_Note.pdf) and the Quest Log (L10_Quest_Log.pdf) with the lesson materials. Families print the Quest Log at 100%, keep it closed in an envelope and find a quest partner (a grown-up or a brother or sister). A family with no printer tells you: arrange another way (for example the make-up slot).
2. Tell the group in the opening and at slide s-quest1: this is Mi's Challenge Quest, a game with Mi. Everyone works on their own paper with a quest partner beside them. It is not a race, a gate may be skipped, and nothing is shown to the group.
3. Start at minute 63 (after the break and the workbook) and finish by minute 83. Children open the Quest Log only when you say. The paper lies flat on the table and is never held up. The quest partner may talk about how to start and ask 'how do you know?', but does not give answers. Cameras may be on or off. Use your own watch out of sight of the camera: no clock, countdown or timer is ever shown to the children.
4. Read each page aloud, one gate at a time, then leave quiet working time: about 2 minutes for page 1, 3 for page 2, 4 for page 3 and 3.5 for page 4. After each page say that the gates are open and Mi swims on, whatever the answers. Say nothing about right or wrong, and read no answers. A middle child needs about 16 minutes in all; a slower child may use the 5 flex minutes.
5. Allowed: fingers, counters, the kit squares, the ruler for gate 7, and talk with the quest partner. Not allowed: a calculator, a grown-up giving answers, looking at the workbook. A child who finishes early turns the Quest Log over and draws quietly. Colouring a

stone on the quest map means I had a go, never I got it right.

6. After page 2, take a one-minute stretch (stand, shake out, sip water). After page 4, show slide s-quest2. Do not collect or look at any Quest Log. A child still writing may use the flex minutes; nothing new is started there. While children work, notice (do not write on the shared screen) how they start: counters, drawing, number line, talking to the partner.

3. After the lesson: photos and your private notes

1. Same day: the grown-up takes ONE clear photo of each of the 4 pages (page flat, taken straight from above, the whole page in view including the 10 cm bar on page 4, and the name on each page) and sends them to the program's business WhatsApp, +65 9880 5962, and to no other number. Never to a child, never to a personal phone, never into the class chat.
2. Within 3 days: open the photos only on the business WhatsApp (never on a personal phone) and write your private notes (section 5) for each child, from the photos and what you saw on camera. Section 4 shows what each answer tells you. For gate 7, the ruler gate, hold the drawn line against the 10 cm bar on page 4: about 6 cm is what we hope to see. If the photo is not clear, ask the grown-up (not the child) for a clearer one, or rely on the answer to part (b). Nothing is written on the child's paper, nothing is added up and no order of children is made. Nothing about the quest goes back to the child or the group except thanks.
3. Write ONE line for each grown-up from your notes (section 7): what we saw, in plain words, and one small next step if it helps. It is sent to the parent only. No marks, no x out of 10, no ranking, no comparison with other children, no prediction.
4. Fill in the group advice sheet (section 6): what several children were still working on, with no names, and what you will do in the first lessons of next term. It is kept for next term's tutor and is never shown to families.
5. Once the line is sent, save the photos to the program's private file store and delete them from the business WhatsApp chat. The program keeps the photos for 90 days after the program ends, then deletes them, as the privacy notice says. Delete each child's notes sheet when the line is sent (keep only the line in the family file, and the group advice sheet). Never forward the photos or save them to a personal phone or any other device.
6. At the start of Term 2 say to the whole group: thank you for playing Mi's Challenge Quest; there is nothing to share and nothing to fear. Never name a child or an answer.

Privacy checklist

- Quest Logs are never held up to the camera and never read out.
- No answer is said aloud during or after the quest. Gates open for everyone, whatever the answers.
- No clock, countdown or timer is shown; nobody is told who finished first or last.
- Photos come to the program's business WhatsApp (+65 9880 5962) only; tutors never use a personal phone; photos are kept 90 days after the program ends, then deleted; the line goes to the parent only; no private message to any child.
- No score, no x out of 10, no ranking, no tally, no comparison between children, not even in your own notes.
- The recording captures the tutor, the shared screen and any voice that is not muted: pause it for the whole quest. The WhatsApp number is never on a slide.

Absent, late joiners, families who prefer not to, upset children

- A child who was away or whose family prefers another time: plays the quest at the make-up slot BEFORE watching the recording (the quest is not on the recording, so no gate is given away). It is a small group or a pair with Mi, with a parent beside the child on camera. The tutor never sits alone with one child on a private call.
- A child who joined late: the family may choose to leave out gates for lessons the child did not attend. Write nothing for those gates; if no gate of an area was taught, the line for the grown-up says nothing about that area.
- A family that does not want the quest: no problem. The child does the Lesson 10 work as usual and the line for the grown-up uses your own notes from the lessons only.
- A child who is upset or stuck: remind the group (open session) that every gate can be skipped and Mi swims round it. Do not single the child out. The grown-up may stop the quest at any time.
- A family with no printer: tell the family to say so a week before and arrange another way (for example the make-up slot, with the Quest Log sent to the family another way). Never ask a child about it in front of the group.

4. Answers and what to look for

Use these to understand what you see on camera and in the photos. Nothing is written on the child's paper, nothing is totalled and no gate is marked right or wrong in front of anyone. A child may write steps in any clear layout. Spelling and neatness do not matter.

Gate 1 (L1) Count the sticks. Fill in the blanks.
Answer: 3 tens and 9 ones; 39 sticks
(a) __ tens and __ ones = 3 tens and 9 ones; (b) There are __ sticks altogether. = 39
A secure answer looks like: 3, 9 and 39 all correct. **Half way there:** Two of the three correct (for example 39 but the tens and ones swapped).
Worked: 3 bundles of ten is 30, and 9 singles makes 39.
Watch for: 39 written as 93.; Counting each bundle as one stick (3 + 9 = 12).

Gate 2 (L2) Fill in the blanks.
Answer: 67; 44
(a) Look at the number line. The arrow is pointing to __. = 67; (b) Counting in 10s: 14, 24, 34, __ = 44
A secure answer looks like: Both correct. **Half way there:** One of the two correct.
Worked: The arrow is just after 66 and just before 68, so it points to 67. In 10s: 14, 24, 34, 44 (the tens digit goes up by 1 and the ones digit stays 4).
Watch for: (a) 66 or 68 (the neighbour).; (b) 35 (counting in ones) or 54 (jumping twice).

Gate 3 (L3) Look at the two cards. Use both cards to make the greatest number you can.
Answer: 83
The greatest number is __. = 83
A secure answer looks like: 83. **Half way there:** 38 written (the other number made from the cards).
Worked: Two numbers can be made: 38 and 83. 83 has 8 tens and 38 has 3 tens, so 83 is the greatest.
Watch for: 38 (the smaller number).; 11 (adding the cards).

Gate 4 (L5) Look at the ten frames. Fill in the missing numbers.
Answer: 12; 42
(a) 7 + 5 = [ ] = 12; (b) 37 + 5 = [ ] = 42
A secure answer looks like: Both correct. **Half way there:** One of the two correct.
Worked: 7 + 3 = 10, and 2 more makes 12. 37 + 3 = 40, and 2 more makes 42.
Watch for: (b) 32 or 312 (joining digits or forgetting the tens).; (a) 11 or 13 from a counting slip.

Gate 5 (L6) Add or subtract.
Answer: 82; 43
(a) 52 + 30 = __ = 82; (b) 74 - 31 = __ = 43
A secure answer looks like: Both correct. **Half way there:** One correct.
Worked: (a) 3 tens more than 52 is 82. (b) 74 - 30 = 44, then 44 - 1 = 43.
Watch for: (a) 55 (adding 3 to the ones) or 522.; (b) 33 (taking away 41) or 44 (taking only the tens).

Gate 6 (L8) Fill in the missing number.
Answer: 6; 20
(a) 8 + 4 = 6 + [ ] = 6; (b) 60 + 10 = 50 + [ ] = 20
A secure answer looks like: Both correct. **Half way there:** One correct.
Worked: (a) 8 + 4 = 12, and 12 is 6 + 6, so the box is 6. (b) 60 + 10 = 70, and 70 is 50 + 20, so the box is 20. Both sides must have the same total.
Watch for: (a) 12 (writing the total of the left side).; (b) 70 (the total instead of the missing number).

Gate 7 (L9) Use your ruler.
Answer: a line 6 cm long; 10 cm
(a) Draw a line 6 cm long. Start at the dot. = A line from the dot, 6 cm long (5.7 to 6.3 cm); (b) A line 4 cm longer than your line is __ cm long. = 10
A secure answer looks like: The line is about 6 cm (5.7 to 6.3 cm; accept 5.5 to 6.5 cm if the photo is not clear) and 10 is written. **Half way there:** Only the line about right, or only the 10 written (a line within 0.5 cm of 6 cm counts if the photo is not clear).
Worked: Put the 0 mark of the ruler on the dot and draw to the 6 mark. A line 4 cm longer is 6 + 4 = 10 cm.
Watch for: Starting the line at 1 on the ruler (a 5 cm line).; (b) 4 (the extra only) or 6.

Gate 8 (L10) Mei has 35 stickers. Ali gives her 10 more stickers. Then she gives 3 stickers to Ben. Write one number sentence for each step. The last number you write is how many stickers Mei has now.

Answer: 35 + 10 = 45; 45 - 3 = 42 (Mei has 42 stickers)

(a) Step 1: .... = 35 + 10 = 45; (b) Step 2: .... = 45 - 3 = 42

A secure answer looks like: Both step sentences have the right operation and numbers (any clear layout) and the second ends with 42.

Half way there: One correct step sentence, or both sentences right with one slip in the arithmetic, or only the answer 42 without steps.

Worked: Step 1: she gets 10 more, so add: 35 + 10 = 45. Step 2: she gives 3 away, so take away: 45 - 3 = 42.

Watch for: 32 (taking the 10 away from 35 and adding 3 or similar).; 48 (adding both numbers).; Using 35 again in Step 2.

Secret cave (optional) (L10) Ali is in a line. There are 5 children in front of him and 3 children behind him. How many children are in the line?

Answer: 9 children (5 + 3 + 1, counting Ali)

There are __ children in the line. = 9

A secure answer looks like: The answer is as above. **Half way there:** -

Worked: 5 in front and 3 behind makes 8, and Ali is in the line as well: 8 + 1 = 9.

Watch for: 8 (leaving Ali out).

5. Tutor private notes (tutor only; one sheet per child; delete when the line is sent)

Child (first name only): ..... Date: ..... Quest partner: .....

Fill this in after the lesson, from the photos and what you saw on camera. Write what the child did and said in your own words. No ticks for right or wrong, no points, nothing added up. The gate numbers are only reminders of where to look.

Gate	From	What I saw (words used, counters or drawing, how the partner helped, anything the child said)
1	L1	
2	L2	
3	L3	
4	L5	
5	L6	
6	L8	
7	L9	
8	L10	
Cave	L10	

Prompt	My notes
Something the child did well (a named strength, for example: explained why 62 is greater than 26 using bundles of ten)	
How they started: counters, drawing, number line, or in their head	
Where they paused or turned to their partner, and what helped	
One next step for home (small and concrete)	
Did they try the secret cave? (a neutral note only)	

The one line for the grown-up (section 7), written from these notes:

.....

.....

6. Group advice for next term (tutor only; one sheet per class; no names)

When all the notes are written, look across them without names. Write what several children were still working on and how you will use the first two lessons of next term (a warm-up, a picture, a game). This page never goes to families.

Area	What several children were still working on (no names)	What I will do in the first two lessons of next term
Numbers, tens and ones Lessons 1 to 4: reading and writing numbers to 100 as tens and ones, number lines, comparing numbers and counting in tens		
Adding and taking away Lessons 5 to 8: adding by making a ten, jumping in tens, taking away tens and ones, and using = as a balance		
Measuring length and solving stories Lessons 9 to 10: drawing and reading lengths in centimetres with a ruler, and reading a story twice and writing one step at a time		

Prompt	My notes
Gates or ideas where several children paused (no names)	
A warm-up or game to bring back in the first lesson of next term	
What worked well about the quest as a game, to keep next time	

7. The one line for the grown-up: what we saw

Mi's Challenge Quest: [Name]
In Mi's Challenge Quest, [Name] [one specific thing you saw, in plain words]. [Optional: one small next step at home.]

One or two sentences at most. Say what the child did (used bundles, drew a line, explained a step), not how many answers were right. Use the lines below as patterns and always add your own detail from your notes.

Area (parent wording)	Things you might have seen (pick the one your notes show; add the detail)	A next step that helps at home
Numbers, tens and ones Lessons 1 to 4. We looked at reading and writing numbers to 100 as tens and ones, number lines, comparing numbers and counting in tens.	• [Name] read numbers as tens and ones using bundles and a number line, and made the greatest number from two cards. • [Name] explained how a number is made of tens and ones, and counted on in tens. • [Name] found the number the arrow showed on a number line and said how they knew.	• Bundle real things in tens (pasta, spoons) and say the tens and ones together. • Count in tens together on a number line or a hundred square.
Adding and taking away Lessons 5 to 8. We looked at adding by making a ten, jumping in tens, taking away tens and ones, and using = as a balance.	• [Name] added by making a ten and used a jump of ten on a number line. • [Name] took away tens and ones and explained each step. • [Name] used the equals sign as a balance and found the missing number.	• Use counters and a number line, with small steps (add 10, take away 10). • Play balance: 8 + 4 = 6 + what? Check with counters.
Measuring length and solving stories Lessons 9 to 10. We looked at drawing and reading lengths in centimetres with a ruler, and reading a story twice and writing one step at a time.	• [Name] used a ruler from the 0 mark and drew a line of the right length. • [Name] read a story twice and wrote one number sentence for each step. • [Name] worked out a longer length from a line already drawn.	• Measure things at home from the 0 mark. • Act out two-step stories with counters, one step at a time.

A filled example (fictional child, Mei)

Mi's Challenge Quest: Mei
In Mi's Challenge Quest, Mei read the sticks as 3 tens and 9 ones, used her ruler from the 0 mark and wrote one number sentence for each step of a story. A good next step: act out two-step stories with counters at home.

This line describes Mei's own thinking in the game. It does not compare children.

Words to avoid in the line: score, mark, out of, rank, top, below average, behind, ahead of, pass, fail, predicted, ready for. No claim about school exams, GEP or any contest, and no mention of other children. The line goes to the parent only. If the child skipped the secret cave or a gate, say nothing about it. If the child tried the secret cave, you may add: "[Name] also had a go at the secret cave."